Akoizo Original Practice - 2027 Rotation Form 1

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) An ampere is…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Joule per coulomb

B. Ohm per meter

C. Coulomb per second

D. Joule per second

Answer: _____

2. (2 points) KCL expresses…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Resistance addition only

B. Charge conservation

C. Equal current in all branches

D. Energy creation

Answer: _____

3. (2 points) A series resistor mainly…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Reverses polarity

B. Limits current

C. Creates charge

D. Guarantees any voltage is safe

Answer: _____

4. (2 points) At fixed R, doubling V changes power by…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. No change

B. A factor of four

C. A factor of two

D. A factor of one half

Answer: _____

5. (2 points) Resistor dissipation is…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. I^2R

B. Always negative

C. Independent of current

D. Equal to charge

Answer: _____

6. (4 points) Why declare the diode model?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

Answer: _____
_____
_____

7. (4 points) Convert 2 Wh to joules.

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

Answer: _____
_____
_____

8. (4 points) Why avoid dependent equations?

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

Answer: _____
_____
_____

9. (4 points) What does a relay separate?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

Answer: _____
_____
_____

10. (4 points) What does the lab omit for AC?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

Answer: _____
_____
_____

11. (2 points) A 7 V ideal battery drives a 3.5 ohm resistor. Find current.

Answer: _____
_____
_____

12. (2 points) A series pair has resistances 3 and 5 ohms. Find equivalent resistance.

Answer: _____
_____
_____

13. (2 points) Two identical 6 ohm resistors are placed in parallel. Find equivalent resistance.

Answer: _____
_____
_____

14. (2 points) An ideal resistor carries 2 A at 7 V. Find power.

Answer: _____
_____
_____

15. (2 points) A node has incoming currents 4 A and 2 A and one outgoing current. Find the outgoing current.

Answer: _____
_____
_____

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

Answer: _____
_____
_____

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

Answer: _____
_____
_____

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

Answer: _____
_____
_____

19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

Answer: _____
_____
_____

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

Answer: _____
_____
_____